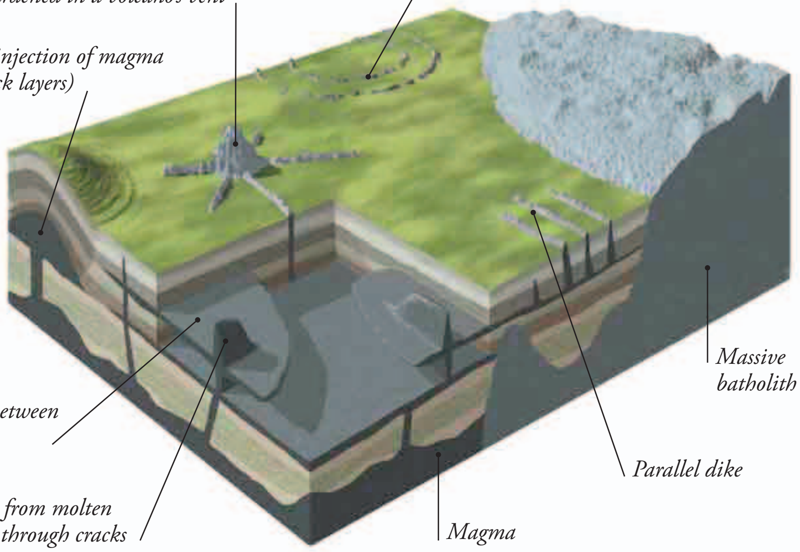
PLUTONIC ROCKS

When magma cools and solidifies deep underground, it forms plutonic rocks. Because it is insulated by surrounding rocks, the cooling can take millions of years, giving the minerals in the molten rock a long time to separate out into crystals. This allows the crystals to grow bigger than they would if the magma cooled quickly, like volcanic lava.

◂ BIG AND SLOW
The giant pink crystals in this granite show that the rock formed deep underground, in a large mass called a pluton.



IGNEOUS INTRUSIONS

Igneous rocks often solidify underground as huge masses called batholiths—igneous intrusions that can be 60 miles (100 km) or more across. But the molten magma can also push up through cracks in the crust to form dikes, or squeeze between rock layers to create sills. These features are all linked together underground, as shown here, but the rocks of dikes and sills have smaller crystals because they have cooled more quickly.

VOLCANIC ROCKS

Rock that erupts from volcanoes cools quickly, so solidified lava has very small, often invisible crystals. The resulting volcanic rocks have their own names. Rhyolite, for example, is a volcanic form of granite. But while these rocks may have no obvious crystal structure, thick lava flows often crack into distinctive forms as they cool and shrink. Basalt, in particular, splits into natural hexagonal columns.

▾ FRACTURED LAVA
The Giant's Causeway in Northern Ireland is a basalt lava flow that erupted around 55 million years ago and split into about 40,000 basalt columns.